Probing Quantum Correlations in Molecular Systems: From Electronic Structure to Tensor Network Methods

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Abstract

This project builds step by step from singular value decomposition and the Schmidt decomposition to a minimal Density Matrix Renormalization Group (DMRG)-style example in NumPy, and uses the Block2 DMRG library alongside standard quantum chemistry methods (Hartree-Fock, CCSD, FCI) to study the H_2 molecule at four bond lengths. The central finding is that both the orbital entanglement entropy and the magnitude of the electron correlation energy increase substantially as the bond is stretched toward dissociation, and that Block2 DMRG reproduces FCI ground-state energies to machine precision throughout. The project was completed independently, building on a B.Sc. in Applied Chemistry with no prior formal coursework in quantum mechanics.

Introduction

Electron correlation plays a central role in determining the accuracy of quantum chemical calculations. While mean-field methods such as Hartree-Fock (HF) provide a computationally efficient starting point, they systematically fail to capture the correlated motion of electrons — a limitation that becomes particularly severe in the dissociation limit of a chemical bond, where the single-determinant picture breaks down.

Tensor network methods, originally developed in the context of condensed matter physics, offer a complementary perspective on this problem. By representing a many-body wavefunction as a chain of low-rank tensors (a Matrix Product State, MPS), these methods make the entanglement structure of a quantum state explicit and computationally tractable. The Density Matrix Renormalization Group (DMRG) algorithm, built on this representation, has become a powerful tool for strongly correlated systems in both condensed matter physics and quantum chemistry.

This project investigates the connection between these two perspectives — electronic structure theory and tensor network methods — using the H_2 molecule as a minimal but instructive test case. Specifically, it addresses the question:

How do orbital entanglement measures evolve as a molecular bond is stretched, and what do they reveal about electronic correlation?

To this end, the project starts from the foundational mathematics of tensor decomposition (singular value decomposition, Schmidt decomposition), implements a minimal DMRG-style local optimization in NumPy that is validated against exact diagonalization on a small spin chain, and then uses the Block2 DMRG library — alongside standard quantum chemistry methods (HF, CCSD, FCI) — to track both the correlation energy and the orbital entanglement entropy of H_2 as a function of bond length.

Methods

Theoretical foundations

The project builds from first principles: complex numbers and linear algebra, the tensor product structure of multi-qubit states, the Singular Value Decomposition (SVD) and its connection to the Schmidt decomposition, the von Neumann entanglement entropy, and the Matrix Product State (MPS) representation of a many-body wavefunction. A minimal DMRG-style variational optimization was implemented from scratch in NumPy, consisting of two local two-site updates with an effective Hamiltonian followed by SVD, and its final energy was compared with exact diagonalization on a three-site Heisenberg spin chain (agreement to numerical precision).

Electronic structure calculations

Quantum chemistry calculations were performed with PySCF: Restricted Hartree-Fock (RHF), Coupled Cluster Singles and Doubles (CCSD), and Full Configuration Interaction (FCI), for the H_2 molecule in the minimal STO-3G basis (two spatial orbitals).

DMRG on H_2

DMRG ground-state calculations were performed using the Block2 library, interfaced with PySCF-derived one- and two-electron integrals, at R = 1.4, 2.2, 3.0, and 4.0 Bohr (4 sweeps, bond dimension 10) — the same geometries used for the FCI and orbital entropy results reported below. A CCSD bond-stretching scan was also performed at R = 1.0, 1.4, 1.8, and 2.2 Bohr; its correlation energies at the two geometries shared with the FCI set (1.4 and 2.2 Bohr) are compared in the Results section.

Orbital entanglement entropy

The single-orbital entanglement entropy was computed at each bond length from the FCI wavefunction. Two equivalent routes were used: (i) the Schmidt (singular value) decomposition of the FCI coefficient matrix, and (ii) the occupation probabilities of each spatial orbital (empty, singly occupied by α or β, doubly occupied) obtained from the one- and two-particle reduced density matrices. For this two-electron, two-orbital system the two routes give the same values, which are reported below.

System and geometries

All calculations were performed on the H_2 molecule in the STO-3G basis, at four H–H bond lengths: 1.4, 2.2, 3.0, and 4.0 Bohr, spanning the equilibrium geometry to a stretched, strongly correlated regime.

Software

Calculations were performed in Python using PySCF 2.14.0 (Hartree-Fock, CCSD, FCI), Block2 0.5.4 (DMRG), and NumPy 2.1.3 (tensor network implementation), run in Google Colab.

Results

Potential energy curves

Figure 1 shows the Hartree-Fock and FCI/DMRG potential energy curves as a function of bond length. The HF and correlated curves coincide near equilibrium but diverge increasingly as the bond is stretched, reflecting the growing importance of electron correlation.

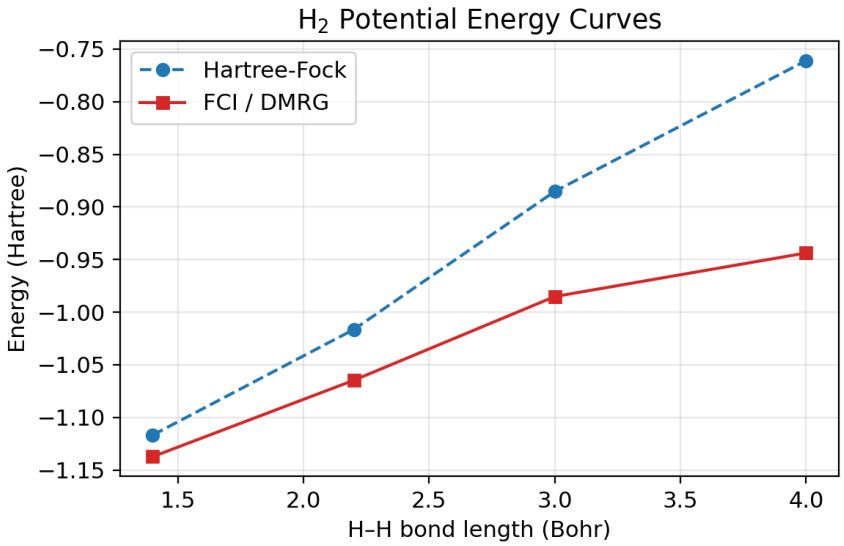


Figure 1. H_2 potential energy curves: Hartree-Fock vs. FCI/DMRG (which agree to machine precision).

Ground-state energy validation

DMRG reproduces the FCI ground-state energies to machine precision at all four tested bond lengths, confirming the correctness of the DMRG implementation and integral interface used.

R (Bohr)	HF (Eh)	FCI (Eh)	DMRG (Eh)	|DMRG−FCI|
1.4	−1.116714	−1.137276	−1.137276	~0
2.2	−1.016486	−1.064739	−1.064739	~0
3.0	−0.885275	−0.985157	−0.985157	~0
4.0	−0.761082	−0.943778	−0.943778	~0

Electron correlation and orbital entanglement

The correlation energy, defined as E_corr = E_FCI − E_HF, grows in magnitude as the bond is stretched. Over the same range, the single-orbital entanglement entropy, computed from the FCI wavefunction, also increases substantially.

R (Bohr)	|Correlation energy| (Eh)	Entropy (bits)
1.4	0.020562	0.098233
2.2	0.048254	0.292052
3.0	0.099882	0.616999
4.0	0.182696	0.906501

As a cross-check, correlation energies from the separate CCSD scan agree with E_FCI − E_HF at the two geometries shared by both sets. For a two-electron system, CCSD is exact in the same basis, so this agreement is expected.

R (Bohr)	|E_corr| from CCSD (Eh)	|E_FCI − E_HF| (Eh)
1.4	0.02056	0.020562
2.2	0.04825	0.048254

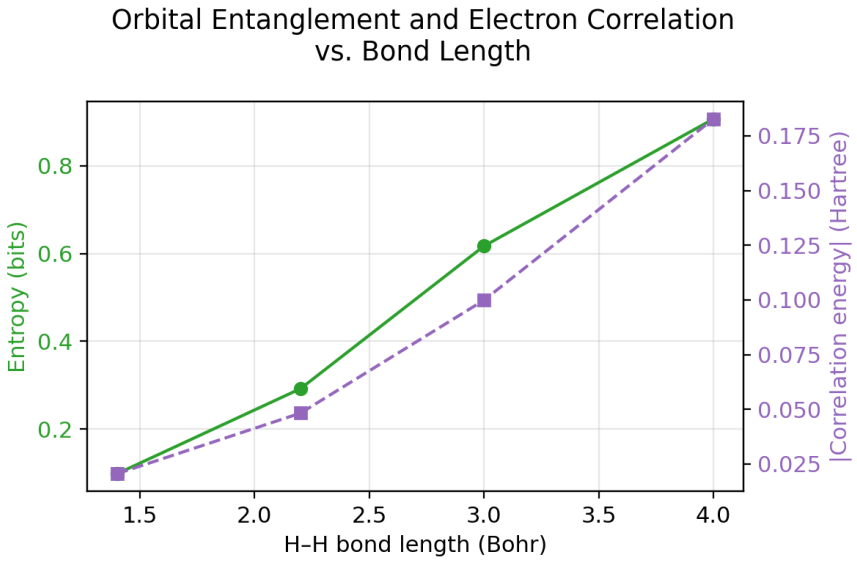


Figure 2. Orbital entanglement entropy and |correlation energy| both increase with bond length.

The two spatial orbitals have equal entropy at every bond length. This follows because the ground state is a pure state of the two-orbital system, and for a pure bipartite state both parts have the same entropy.

Discussion

The central result of this project is that both the magnitude of the electron correlation energy and the orbital entanglement entropy increase substantially as the H–H bond is stretched toward dissociation — the entropy nearly an order of magnitude, from 0.098 to 0.907 bits, over the range R = 1.4 to 4.0 Bohr. This parallel growth is consistent with the physical picture that a single Hartree-Fock determinant becomes an increasingly poor description of the true electronic state as static correlation becomes dominant near dissociation, and that this breakdown is directly reflected in the growing entanglement between orbitals.

It is worth noting that while correlation energy and orbital entanglement entropy both increase together in this dataset, they are not equivalent quantities: the former is an energetic measure derived from a specific pair of methods (FCI and HF), while the latter is an information-theoretic property of the wavefunction itself. Their parallel behavior here is a meaningful but not a logically necessary relationship.

This agreement between DMRG and FCI across all tested geometries validates the DMRG implementation and its integration with the PySCF-derived molecular integrals via Block2.

Limitations

- H_2 in a minimal (STO-3G) basis is a two-orbital system, used here as a minimal, instructive test case rather than a large-scale molecular benchmark.

- Because FCI is computationally inexpensive for this two-orbital problem, the close agreement between DMRG and FCI validates the implementation but does not demonstrate a computational advantage of DMRG over exact diagonalization; that advantage becomes relevant for larger active spaces.
- The DMRG implementation built from scratch (used for the three-site Heisenberg validation) performs two local optimization steps rather than a complete, iterative left-right/right-left sweep to convergence. All molecular DMRG results (H_2) were obtained with Block2, which performs full sweeps (4 sweeps, bond dimension 10).
- The three-site validation uses an exact bond dimension, so it tests the correctness of the implementation, not the efficiency of truncated DMRG.
- The single-orbital entropy reported here captures one aspect of orbital correlation and does not describe every form of entanglement present in the molecular wavefunction.
- Results are reported for four bond lengths in one basis set; extending to larger basis sets, additional geometries, and other molecules is a natural next step.

Conclusion

This project built a connection between electronic structure theory and tensor network methods, starting from the mathematical foundations of entanglement and tensor decomposition and culminating in a DMRG study of electron correlation in H_2. The central findings are that the magnitude of the FCI–HF correlation energy and the single-orbital entanglement entropy (computed from the FCI wavefunction) both increase substantially over the sampled bond-stretching range. Separately, Block2 DMRG reproduces the FCI ground-state energies to machine precision at all four tested geometries, validating the DMRG implementation and its integration with the PySCF-derived molecular integrals.

Code, notebooks, and full results are available at: github.com/atoosa-taheri/quantum-correlations-mps